

Background

In 2018 292.4 million tons of municipal solid waste were generated. That is **4.9 pounds per person per day**, the greatest per-capita value ever recorded. Despite efforts to recycle, compost, and combust for energy reclamation, 146 million tons of this waste ended up in landfills. The best method for waste management is always to decrease waste generation. This will require **people to make changes in their habits**, which is very difficult to enact or maintain without timely and relevant feedback.

Objective

Our capstone project aims **to help environmentally conscious individuals** reduce their household waste production. Our goal has been to create a system which can quantify the impact of any changes the user makes in their waste habits. This system will **track and log the cumulative weight** of waste added to a trash can over time.

Design

The guiding principle for the development of our system was that it should be **simple and able to integrate** with a wide range of existing trash cans. As can be seen in the sketches below, there were originally three main components of the design:

1. **Weighing platform** with stress sensors on the bottom to measure the total weight of the trash can, similar to a bathroom scale
2. **Switch** or other sensor to detect trash can interaction (*not included in the final design*)
3. **Micro-controller** with display and buttons for user input, logs data and manages trash bag replacements

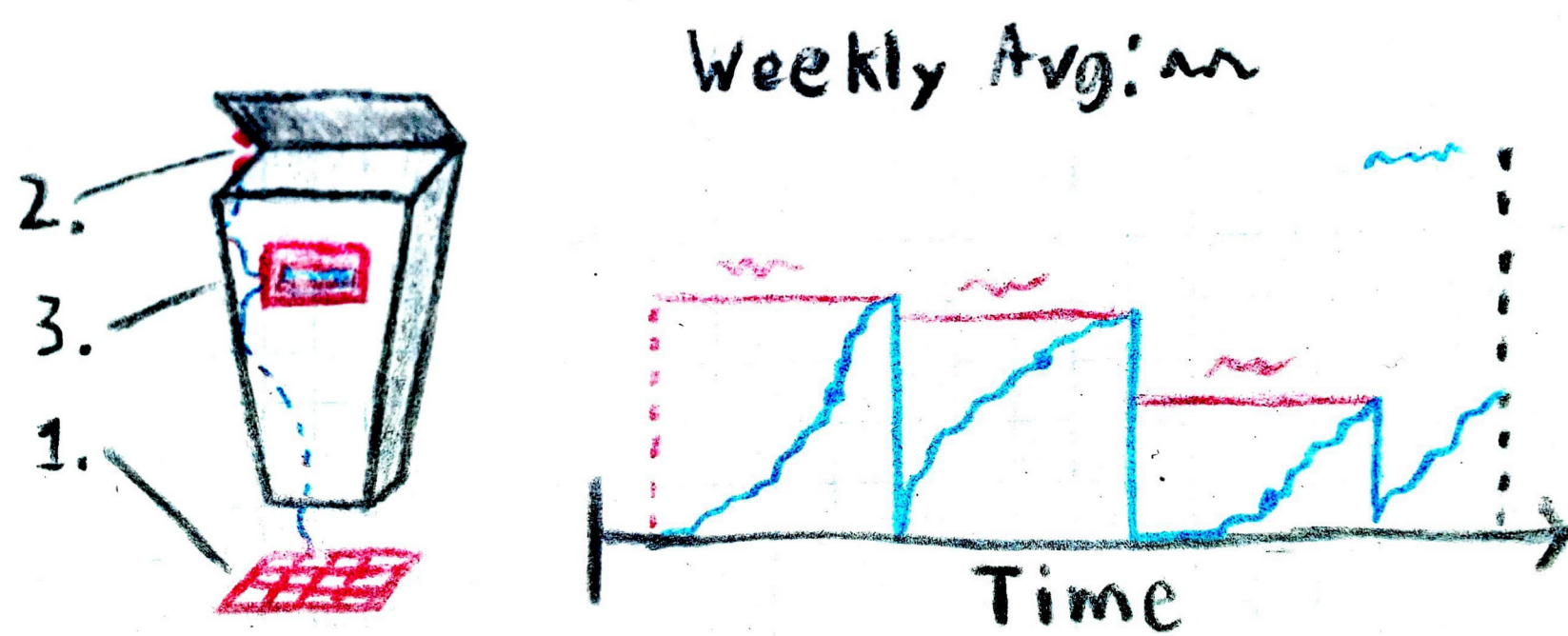


Fig: System diagram

Fig: Imagining of data visualization

Fig: The garbage team

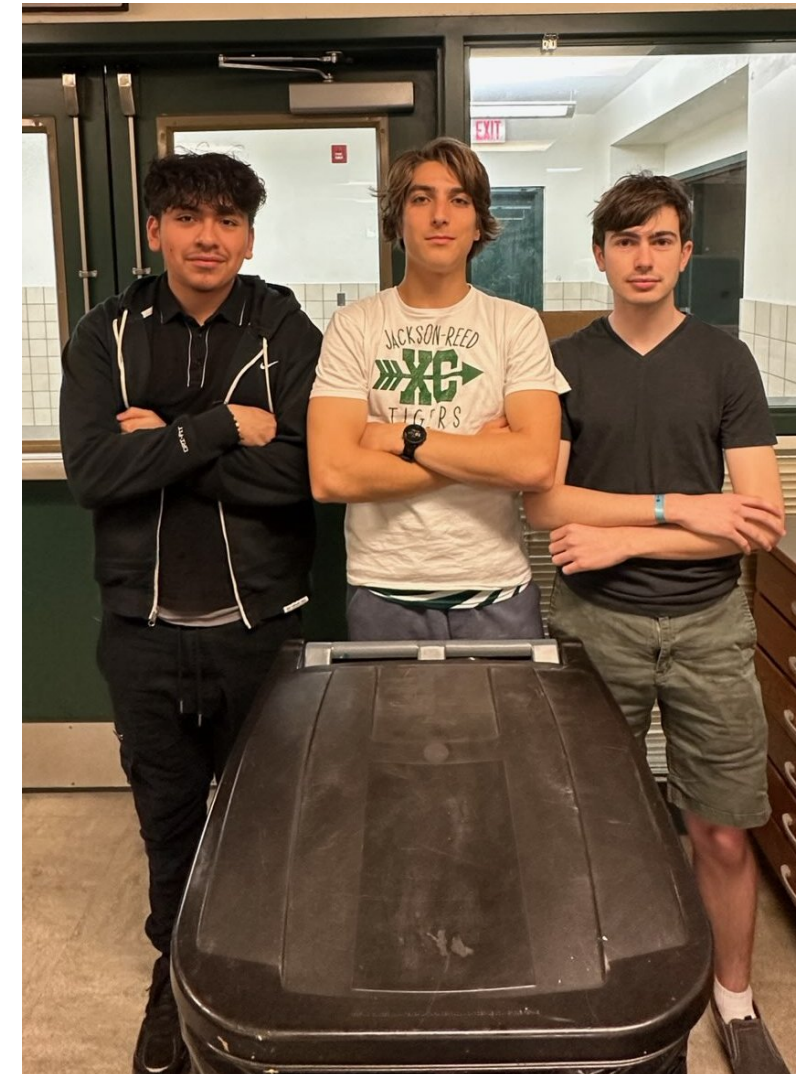
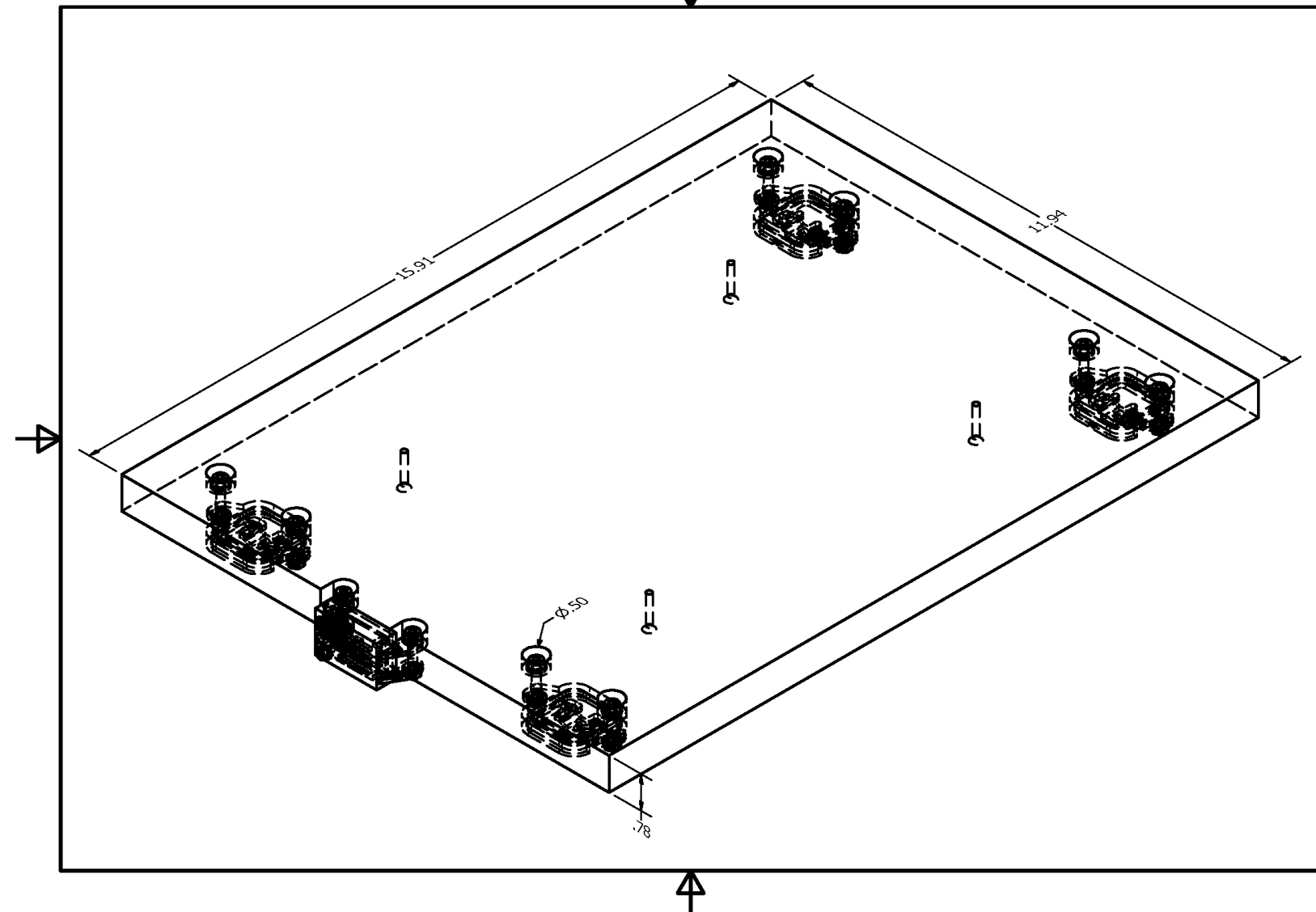


Fig: Inventor model of weighing platform



Prototyping

Our system is **centered around an Arduino Uno Wi-Fi REV2**, which is connected to all the inputs and outputs to log data to an SD card. The SD card reader and a RTC are both integrated with the **Adafruit Data Logging Shield**, which additionally provides prototype board space for other components. The LCD panel, input buttons (which are on a separate prototype board due to limited space), indicator light, and JST connector for the weighing platform cable **are all soldered here**.

The weighing platform consists **of four stress sensors wired in a Wheatstone bridge** and mounted on the underside of a plywood board. The mounts for the HX711 board and the stress sensors were all 3D printed based on the Autodesk Inventor model of the entire weighing platform assembly, using bolts as fasteners.

A plethora of woodworking tools were used to **cut, drill, countersink, and cut slots in the platform board**, including the school's CNC router. A FDM 3D printer was used for the stress sensor mounts, and a resin printer was used for the electronics casings. **Much trial and error was needed** in order get certain components to fit correctly, as well as to develop the program to execute the operational logic correctly.

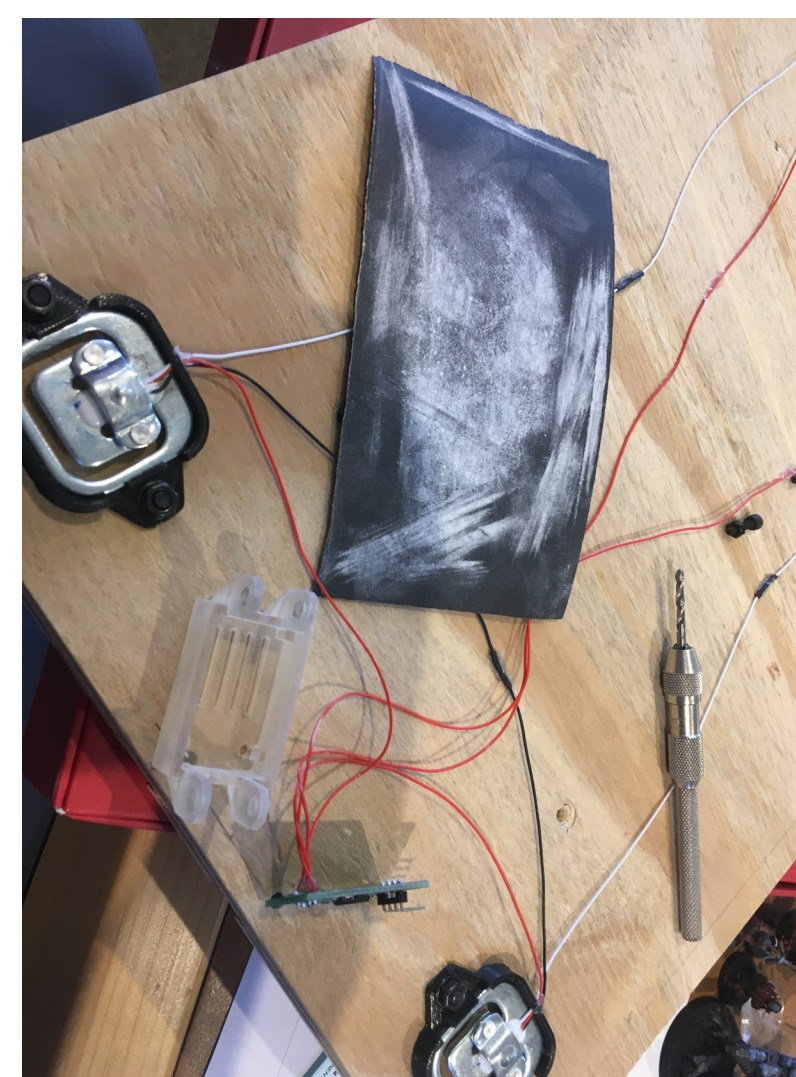


Fig: Fitting the HX711 mount

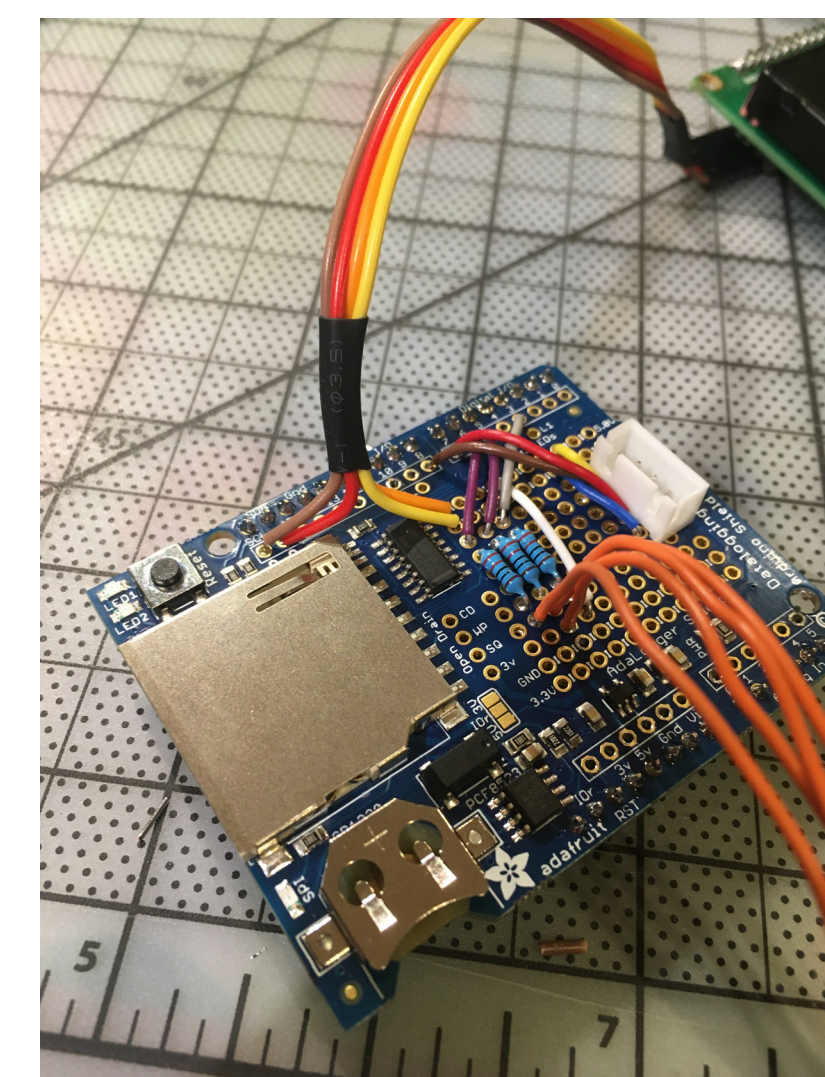


Fig: Completed wiring Arduino

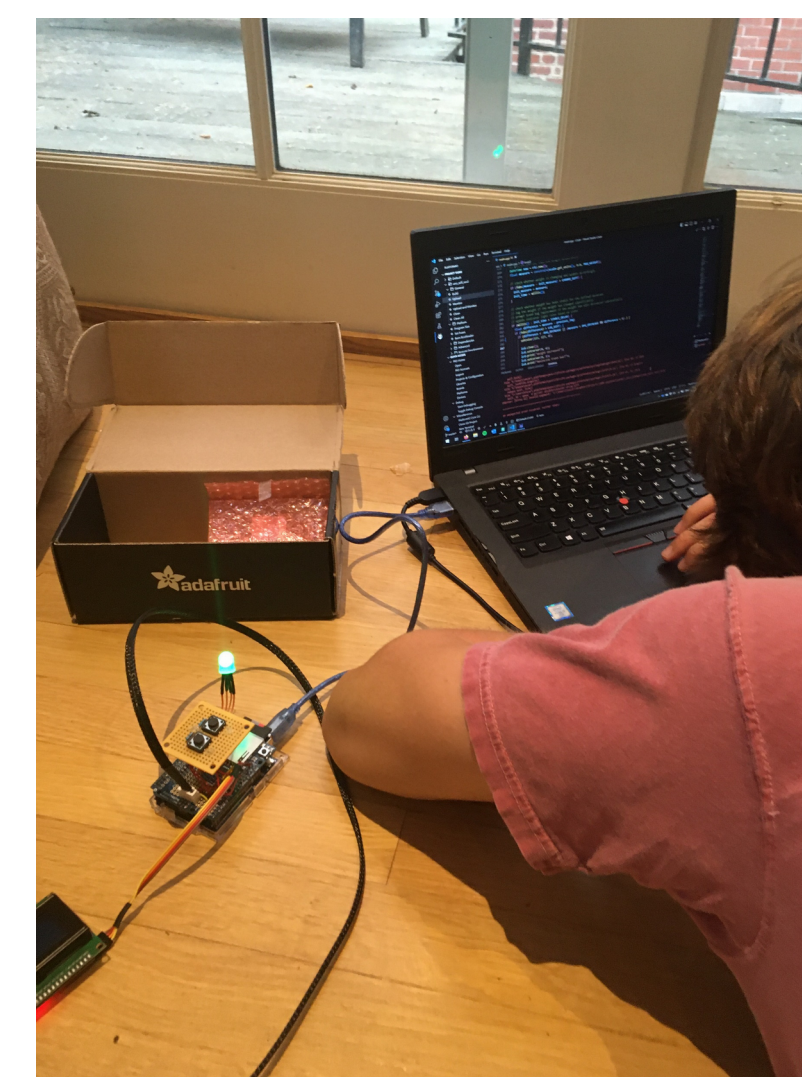


Fig: Bug fixes during testing

Implementation

The Arduino program operates and logs data **based on the following logic**:

- After initiating the sensors and outputs enter the main loop, displaying the time and current weight
- If either button is pressed, ask the user whether they want to tare the weight; this will not log a trash bag
- If the weight has been **stable to within ± 0.01 kg** and has **changed more than 0.02 kg** from the last measurement, log the weight
- If the weight is to be logged and has **decreased by more than 0.10 kg**, ask the user whether to log a trash bag
- Tare the weight every time a trash bag is logged; log a measurement of 0.00 kg every time the weight is tared

Testing

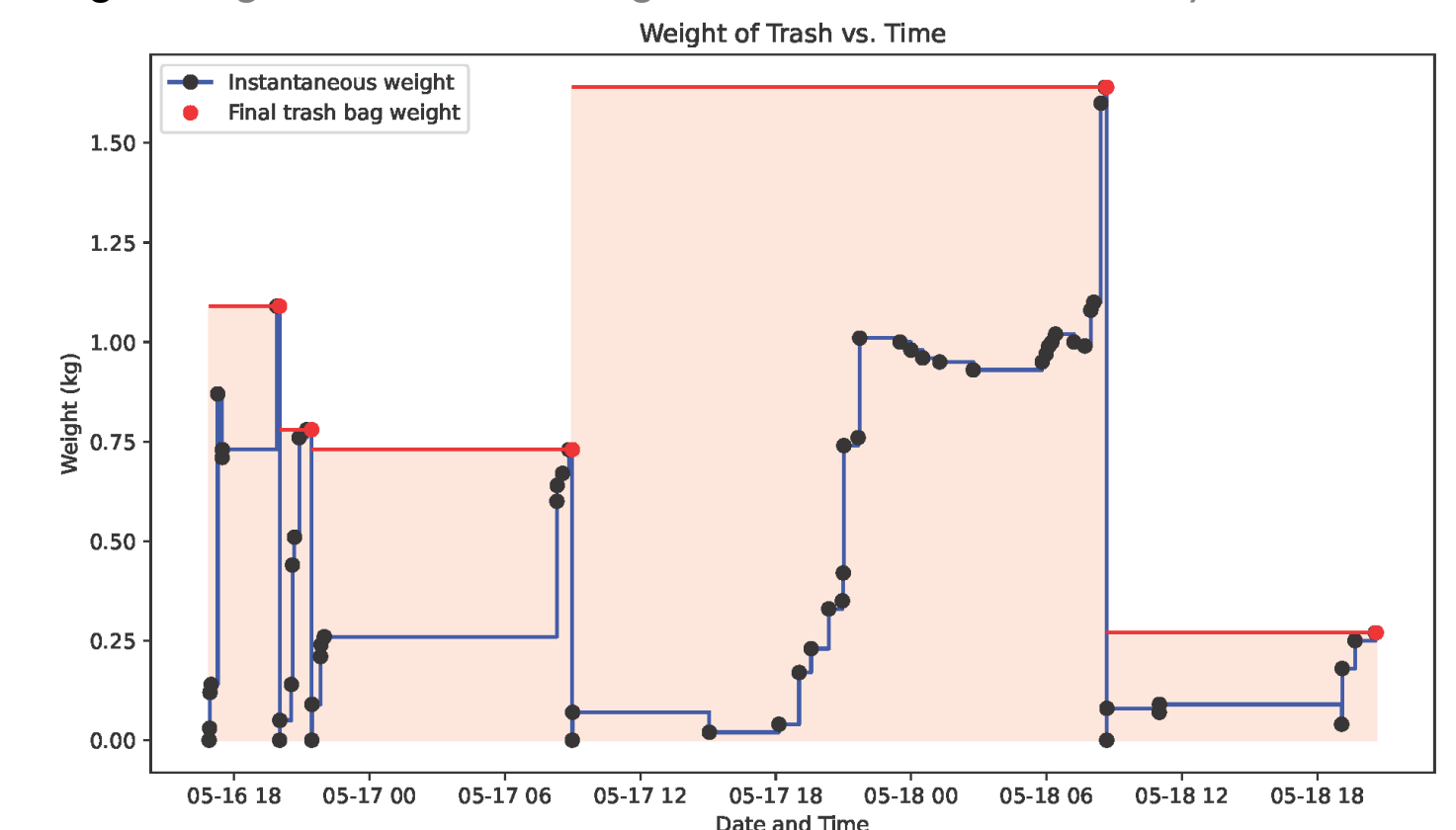
Our testing was severely limited in duration and scope due to many factors, primarily delays in prototype development. Thus, these processes focused more on the **functionality of the system as intended** as opposed to the effectiveness of the system at decreasing household waste over time. The system was set up with a spare trash can, and **paper waste was added or removed** at irregular intervals and in varying quantities.

Through this process, **many flaws in the user interface** and operation of the program were identified and addressed. A loose connection on the buttons' prototype board was identified and fixed. Most importantly, this testing demonstrated that the data gathered **can be used to visually represent** the amount of waste produced over time. The Python program which generated the graph below calculated a total of **4.51 kg** of waste was added to the trash can over the course of three days.

Fig: Temporary testing setup



Fig: Data gathered from testing over the course of three days



Next Steps

The next steps in our development process are to refine the prototype, with an emphasis on reliability, in order to **execute long term testing**. Though it is functional, our system will occasionally freeze and require a restart or incorrectly detect a trash bag replacement. Once these issues have been ironed out, we will be able to assess its actual effects on a user's waste habits over time. We also plan to **explore IoT functionality**, particularly for app integration, such that the user does not have to remove the SD card to load and analyze data. This will allow for features such as **goal setting and deeper analytics**.